



Rumaisa Habib

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Art by [Haider Naveed](#)

Research Overview

My research focuses on understanding the web at a global scale. I combine large-scale measurement and systems building to study how the web works in practice and who it actually serves, especially outside Western contexts. I aim to replace informal intuitions about the web with rigorous evidence that can inform both system design and policy.

Education

Ph.D. Candidate in Computer Science, Stanford University — Expected 2028

M.S. in Computer Science, Stanford University — 2026

B.S. in Computer Science, LUMS — 2023

→ Senior Thesis: *A Framework for Improving Web Affordability and Inclusiveness*

Publications

- [1] ["Stop Trying to Make Fetch Happen": Towards a Modality-Aware Understanding of Unnecessary Resources on the Web](#)
Rumaisa Habib, Balaji Balachandran, Nurullah Demir, Zakir Durumeric
ACM Internet Measurement Conference (IMC '26)
- [2] [On IP Addresses as Identifiers of Internet Users and Services](#)
Rumaisa Habib, Sudheesh Singanamalla, Marwan Fayed, Zakir Durumeric
Research Conference on Communications, Information and Internet Policy (TPRC '26)
- [3] [Don't Get Caught, Keep Your Onions in a Vault](#)
Humza Ikram*, **Rumaisa Habib***, Muaz Ali*, Zartash Afzal Uzmi (*joint first authors)
New Ideas in Networked Systems (NINEs '26)
- [4] [Formalizing Dependence of Web Infrastructure](#)
Rumaisa Habib, Kimberly Ruth, Gautam Akiwate, Zakir Durumeric
ACM Conference of the Special Interest Group on Data Communication (SIGCOMM '25)
IRTF Applied Networking Research Prize
- [5] [Blade: Towards Scalable Source Code Debloating](#)
Muaz Ali, **Rumaisa Habib**, Ashish Gehani, Sazzadur Rahaman, Zartash Afzal Uzmi

IEEE Secure Development Conference (SecDev '23)

- [6] [A Framework for Improving Web Affordability and Inclusiveness](#)

Rumaisa Habib*, Sarah Tanveer*, Aimen Inam, Haseeb Ahmed, Ayesha Ali, Zartash Afzal Uzmi, Zafar Ayyub Qazi, Ihsan Ayyub Qazi (*joint first authors)

ACM Conference of the Special Interest Group on Data Communication (SIGCOMM '23)

- [7] [A First Look at Public Service Websites from the Affordability Lens](#)

Rumaisa Habib, Aimen Inam, Ayesha Ali, Ihsan Ayyub Qazi, Zafar Ayyub Qazi

ACM Web Conference (TheWebConf '23)

- [8] [Rethinking Web for Affordability and Inclusion](#)

Ihsan Ayyub Qazi, Zafar Ayyub Qazi, Ayesha Ali, Muhammad Abdullah, **Rumaisa Habib**

ACM Workshop on Hot Topics in Networks (HotNets '21)

Preprints

[Understanding the "Airport" Censorship Circumvention Ecosystem in China](#)

Rumaisa Habib*, Mingshi Wu*, Shiva Shahandeh, Min Ni, Eric Wustrow, Zakir Durumeric (*joint first authors)

arXiv preprint.

Reports & Blog Posts

- **Author**, *The 2025 Web Almanac* (HTTP Archive) — "Chapter 8: Privacy" (January 2026) by Rumaisa Habib, Vinod Tiwari, Nurullah Demir [\[Link\]](#)
- **Guest Author**, Internet Society Pulse — "Characterizing Internet Centralization vs Regionalization" (December 2025) [\[Link\]](#)

Grants, Fellowships & Honors

- **IRTF Applied Networking Research Prize** (2026)
- **David Cheriton Stanford Graduate Fellowship**, Stanford University — 2023–Present
- **Students as Co-Researchers Grant**, LUMS — 2023
- **Graduated with High Distinction**, LUMS — 2023
- **Merit Scholarship**, LUMS — 2020–2023
- **Dean's Honor List**, LUMS — 2019–2022

Teaching Experience

- **Course Assistant**, Department of Computer Science, Stanford University
 - ◆ CS356: Topics in Computer and Network Security (Fall 2026)
- **Teaching Assistant**, Department of Computer Science, LUMS
 - ◆ CS678/EE678: Topics in Internet Research (Spring 2023)
 - ◆ CS334/EE334: Principles and Techniques of Data Science (Fall 2022)

- ◆ CS3812: Introduction to Blockchain: Technology and Applications (Spring 2022)
- ◆ CS471/EE471/CS573: Computer Networks: Principles & Practices (Fall 2021)

Conference Presentations & Invited Talks

- **On IP Addresses as Identifiers of Internet Users and Services**
 - ◆ TPRC '54, September 2026, Washington DC, USA
- **Formalizing Dependence of Web Infrastructure**
 - ◆ Stanford Security Lunch, July 2026, Stanford, CA, USA
 - ◆ IRTF Open, IETF-126, July 2026, Vienna, Austria
- **A Framework for Improving Web Affordability and Inclusiveness**
 - ◆ IAB BIAS Workshop, January 2024, Virtual
 - ◆ UW CHANGE Seminar, November 2023, Virtual, University of Washington, USA
 - ◆ SIGCOMM '23, September 2023, Columbia University, NY, USA
- **A First Look at Public Service Websites from the Affordability Lens**
 - ◆ TheWebConf'23, May 2023, UT Austin, TX, USA

Service & Professional Affiliations

Departmental & University Service

- **Organizer**, Weekly Security Seminar, Department of Computer Science, Stanford University — 2024–2025
- **IT & Design Director**, LUMS Women in Computing (LWiC) — 2021–2022
- **Research Lead**, Blockchain Research Group, LUMS — Summer 2022

External Review

- USENIX Security 2026
- Internet Measurement Conference (IMC) 2024

Press/Interviews

- **Wall Street Journal** — *"FBI Shuts Down Sprawling China-Linked Hacking Network"* (August 2026) [\[Link\]](#)
- **PPC Land** — *"Stanford study finds 5% of IP addresses send 55% of all web requests"* (August 2026) [\[Link\]](#)
- **Tech Times** — *"IETF 126 Closes Out in Vienna: DPI, Fingerprinting, Prize & BGP Security Push"* (July 2026) [\[Link\]](#)